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Miao Miao

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RESEARCH INTERESTS

Advancing **cybersecurity** and **software engineering** through systematic evaluation and **AI-driven** techniques to build more **secure software** and **intelligent testing and analysis tools**.

EDUCATION

Doctor of Philosophy, Software Engineering — *Jan 2023 – Summer 2027*
The University of Texas at Dallas, Richardson, Texas, USA — *(Expected Graduation)*
Advisor: Dr. Shiyi Wei — GPA: 3.97

Master of Science, Software Engineering — *Aug 2021 - Dec 2022*
The University of Texas at Dallas, Richardson, Texas, USA — GPA: 3.97

Bachelor of Engineering, Computer Science and Technology — *Sep 2014 - June 2018*
The Xi'an University of Finance and Economics, Xi'an, China — GPA: 3.59

AWARDS

- **Invited to the ACM Student Research Competition Grand Finals, 2025–2026 🔗.**
- **Winner of ICSE 2025 ACM student research competition (1st place) 🔗.**
- **Mary and Richard Templeton Graduate Fellowship in 2025.**
- **The ACM SIGSOFT CAPS Travel Award for ICSE 2025.**
- **The Jonsson School Best Teaching Assistant Award in 2024.**

PUBLICATIONS

** ICSE and ISSTA are top-tier conferences in Software Engineering, while TOSEM is among the field's leading journals.*

- **Towards the Automated Identification of Data Constraints in Software Documents**
Ying Zhou, **Miao Miao**, Vlad Birsan, Oscar Chaparro, Shiyi Wei, and Andrian Marcus.
Requirements Engineering (RE), vol. 31, no. 2, pp. 22, Springer London, 2026.

- **Static Analysis Traces can Help Dynamic Symbolic Execution: a Replication Study**
Sriteja Kummita, Fabian Schiebel, Eric Bodden, **Miao Miao**, and Shiyi Wei.
In the 2026 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), pp. 981–991, IEEE, 2026.

- **Visualization Task Taxonomy to Understand the Fuzzing Internals**
Sriteja Kummita, **Miao Miao**, Eric Bodden, and Shiyi Wei.
ACM Transactions on Software Engineering and Methodology (TOSEM), vol. 35, no. 3, pp. 1–27, ACM, 2026.

- **SeedUI: Understanding Initial Seeds in Fuzzing**
Sriteja Kummita, Eric Bodden, **Miao Miao**, and Shiyi Wei.
In the 40th IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW), pp. 181–185, IEEE, 2025.

- **Program Feature-based Benchmarking for Fuzz Testing**

Miao Miao, Sriteja Kummita, Eric Bodden, and Shiyi Wei.
In the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), pp. 527–549, ACM, 2025.
</> Code Repository

- **Program Feature-based Fuzzing Benchmarking**

Miao Miao.
In the 47th IEEE/ACM International Conference on Software Engineering, Companion Proceedings (ICSE-Companion), pp. 232–234, IEEE, 2025.
1st Place, ICSE 2025 ACM Student Research Competition.

- **An Extensive Empirical Study of Nondeterministic Behavior in Static Analysis Tools**

Miao Miao, Austin Mordahl, Dakota Soles, Alice Beideck, and Shiyi Wei.
In the 47th IEEE/ACM International Conference on Software Engineering (ICSE), pp. 1064–1076, IEEE, 2025.
</> Code Repository

- **Visualization Task Taxonomy to Understand the Fuzzing Internals (Registered Report)**

Sriteja Kummita, **Miao Miao**, Eric Bodden, and Shiyi Wei.
In the Proceedings of the 3rd ACM International Fuzzing Workshop (FUZZING), pp. 13–22, ACM, 2024.

- **ECSTATIC: Automatic Configuration-Aware Testing and Debugging of Static Analysis Tools**

Austin Mordahl, Dakota Soles, **Miao Miao**, Zenong Zhang, and Shiyi Wei.
In the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), Tool Demonstration Track, pp. 1479–1482, ACM, 2023.
</> Code Repository

Papers Under Review

- **Mining Repositories to Understand Common Challenges with Static Analysis Frameworks**

Miao Miao, Eric Su Zhang, Fabian Schiebel, Sriteja Kummita, Kadiray Karakaya, Mohamed Soliman, Eric Bodden, and Shiyi Wei.
Under review at the 24th IEEE/ACM International Conference on Mining Software Repositories (MSR), 2027.

- **An Experience Report on Teaching Software Accessibility From a Testing Perspective**

Cynthia Marie Rivera Sánchez, **Miao Miao**, Vlad Birsan, and Shiyi Wei.
Under review at the 49th IEEE/ACM International Conference on Software Engineering, Software Engineering Education and Training Track (ICSE-SEET), 2027.

- **Why Do Fuzzers Get Stuck? Systematically Discovering Fuzzing Challenges in Real Programs**

Miao Miao, Eric Bodden, and Shiyi Wei.
Under review at the 49th IEEE/ACM International Conference on Software Engineering, New Ideas and Emerging Results Track (ICSE-NIER), 2027.

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant

** Contributed to curriculum development and design of projects and assignments; Provided video and in-class tutorials to guide students through project implementation.*

- CS/SE 6356: Software Maintenance Evolution and Re-Engineering (Spring 2023)
- CS 4386: Compiler Design (Fall 2023)
- CS 6353: Compiler Construction (Spring 2024)

Received the Jonsson School Best Teaching Assistant Award in 2024.

Research Mentoring

- Mentored four high school students and three undergraduate students in the UTD K-12 Summer Research Program and Clarks Summer Research Program (Summer 2023, Summer 2024, Summer 2025).
- Advised high school student Eric Su Zhang (now at Washington University in St. Louis) on an independent research project, resulting in a co-authored paper under review at MSR 2027.
- Advised undergraduate student Alice Beideck on an independent research project, resulting in a co-authored peer-reviewed publication at ICSE 2025.
- Advised undergraduate students Alen Kurian Jo and Alice Beideck through the UTD Summer Platform for Undergraduate Research (SPUR), guiding research idea development, poster design, and oral presentation preparation for the student research competition.

INDUSTRY EXPERIENCE

iOS Engineering Intern, Tinder Inc. *May 2022 – Aug 2022*

- Developed iOS localization tooling that automated multi-locale releases and improved translation efficiency.

iOS Developer, LotusFlare Inc. *Aug 2019 – June 2021*

- Contributed to the development and maintenance of two production mobile apps (mWell PH, myIM3).

Software Engineer, KA Software *Apr 2018 – Apr 2019*

- Implemented EDI solutions (X12/EDIFACT) and AS2/SFTP integrations to automate B2B transactions.

SERVICES

- Artifact Evaluation Committee member: the 41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026).
- Artifact Evaluation Committee member: the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026).
- Shadow Program Committee member: the 48th IEEE/ACM International Conference on Software Engineering (ICSE 2026).
- Artifact Evaluation Committee member: the 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2025).
- Artifact Evaluation Committee member: the 46th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI 2025).
- Junior Program Committee member: the 22nd International Conference on Mining Software Repositories (MSR 2025).
- Student Volunteer: the 47th IEEE/ACM International Conference on Software Engineering (ICSE 2025).
- Served as a student research competition judge for the UTD Summer Platform for Undergraduate Research (SPUR) (Summer 2023, Summer 2024, Summer 2025).
- Student Volunteer: the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2023).
- Sub-review: ICSE 2027, FSE 2027, ICSE 2026, ISSTA 2026, FSE 2026, USENIX 2026, CCS 2025, USENIX 2025, ISSTA 2025, ICSE 2025, FSE 2025, MSR 2025, USENIX 2024, ISSTA 2024, ICSE 2024, FSE 2024, ASE 2024, FSE 2023, ISSTA 2023, SecDev 2023, ICSE 2023.

REFERENCES

- **Dr. Shiyi Wei**
  Associate Professor
  Dept. of Computer Science, University of Texas at Dallas
  swei@utdallas.edu

- **Dr. Eric Bodden**
  Professor
  Heinz Nixdorf Institute, Paderborn University
  Co-Director
  Fraunhofer Institute for Mechatronic Systems Design
  eric.bodden@uni-paderborn.de

- **Dr. Andrian Marcus**
  Professor
  Dept. of Computer Science, George Mason University
  amarcus7@gmu.edu